SUPPLY CHAIN VISIBILITY SYSTEM WITH OPTIMIZATION ANALYTICS

PROJECT DOCUMENTATION

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1. Introduction

The **Supply Chain Visibility System with Optimization Analytics** is a Power BI-based project that analyzes inventory, delivery performance, transportation costs, supplier reliability, disruption risks, and warehouse operations.

The project uses Power BI Desktop for dashboards, Power Query for data cleaning and transformation, and DAX for calculating key performance indicators (KPIs). Through interactive visualizations and data modelling, the system helps monitor supply chain performance, identify operational bottlenecks, and support data-driven decision-making.

2. Project Objectives

- Develop an interactive Power BI-based supply chain analytics solution from transaction-level data.
- Establish inventory visibility and inventory-turnover diagnostics.
- Measure delivery workload, on-time performance and delay severity.
- Analyze transportation cost, shipment volume and shipping-mode performance.
- Evaluate supplier quality, reliability and disruption exposure.
- Use slicers, drill-downs, time filters and diagnostic visuals for interactive analysis.
- Progressively extend the dashboard from inventory visibility to delivery analytics and transportation/supplier performance.

3. Project Description

The Supply Chain Visibility System with Optimization Analytics is a Power BI-based analytical project developed to provide end-to-end visibility into inventory, orders, suppliers, transportation, delivery performance, disruptions, and warehouse operations. The project converts supply-chain transaction data into an interactive reporting environment in which users can move from high-level KPIs to detailed operational diagnostics.

The project is needed because supply-chain activity is distributed across several operational dimensions. Looking only at order volume or delivery delay does not provide enough information to understand where operational problems originate. The dashboard therefore combines inventory, logistics, supplier, transportation, and warehouse indicators so that workload, service performance, risk, cost, and facility efficiency can be examined together.

The project is implemented through four milestones:

Milestone 1 establishes the data model and baseline supply-chain visibility and inventory analysis.

Milestone 2 extends the analysis to inventory movement, delivery performance, SLA compliance, and delay behaviour.

Milestone 3 introduces transportation-cost estimation, supplier performance, shipment efficiency, and disruption-risk analysis.

Milestone 4 adds warehouse-level analytics covering capacity, utilization, operational cost, picking accuracy, and facility performance optimization.

Together, the four milestones transform the initial transaction dataset into a connected analytical system covering the major operational layers of the supply chain.

4. Tools Used

The project is implemented using Power BI and its associated data-preparation and analytical features. The following tools and technologies form the main technical foundation of the project.

The tools used in the project serve different stages of the workflow, from preparing the raw dataset to building the analytical model and presenting the final results.

Tool / Technology	Use in Project
Power BI Desktop	Dashboard development, data modeling, visualizations, slicers, drill-downs, page navigation and interactive analysis.
Power Query	ETL, cleaning, standardization, transformation, type conversion and dimension preparation.

DAX	Calculated columns, measures, KPIs, date calculations and analytical logic.
Kaggle Supply-Chain Dataset	Source transactional and operational supply-chain data used for the dashboard analysis.
Star Schema / Power BI Model	Fact-and-dimension structure for consistent relationships and filter propagation.

The combination of these tools allows the project to move from raw transaction records to a structured analytical model and finally to an interactive decision-support dashboard.

5. Milestone 1 – Dashboard Explanation

Supply Chain Visibility & Inventory Turnover

Milestone 1 establishes the foundation of the project through the Supply Chain Visibility and Inventory Turnover dashboard. The purpose of this milestone is to provide an overall picture of supply-chain activity before moving into more specialized delivery and transportation analysis. The dashboard focuses on order volume, quantity ordered, order value, inventory-efficiency indicators, product categories, shipping modes and delay behaviour.

The supplied Milestone 1 documentation identifies important transaction and operational fields including Order_ID, Order_Date, Quantity_Ordered, Order_Value_USD, Shipping_Mode, Product_Category, Supplier_ID, Buyer_ID, Delay_Days, Historical_Disruption_Count, Disruption_Type, Supplier_Reliability_Score and Supply_Risk_Flag. These fields provide the foundation for the dashboard's different analytical views.

4.1 Data Preparation and Model

The data is prepared before visualization so that the fields can be used consistently across the report. The preparation process includes handling unusable records, standardizing date fields, converting numerical and categorical fields to suitable data types, checking duplicate records and preparing dimension tables. The project uses a star-schema approach in which the central transaction data is connected to descriptive dimensions such as calendar, product, logistics, supplier and buyer.

The Milestone 1 documentation identifies the star schema as the structural core of the report and emphasizes one-to-many relationships from dimensions to the fact table so that selections made through dates, modes, categories, suppliers or buyers propagate correctly through the dashboard.

Measure	DAX Formula	Purpose
Total Orders	`Total_Orders = DISTINCTCOUNT('supply_chain_resilience'[Order_ID])`	Counts the unique orders in the current filter context.
Total Units Ordered	`Total_Units_Ordered = SUM('supply_chain_resilience'[Quantity_Ordered])`	Calculates the total number of units ordered.
Total Order Value	`Total_Order_Value = SUM('supply_chain_resilience'[Order_Value_USD])`	Calculates the total monetary value of orders.
Inventory Turnover	`Inventory_Turnover = DIVIDE([Total_Order_Value], [Total_Order_Value] * 0.25, 0)`	Estimates inventory turnover using the defined inventory assumption.
Days of Inventory	`Days_of_Inventory = DIVIDE(([Total_Order_Value] * 0.25) * 365, [Total_Order_Value], 0)`	Estimates the number of inventory days.
Dynamic Category Turnover	`Dynamic_Category_Turnover = DIVIDE(CatUnits, CatHoldEst * 0.2, 0)`	Provides a category-level turnover indicator using units and delay-based holding estimation.

4.2 Dashboard Components

The dashboard components are arranged so that each visual contributes a specific part of the overall analysis. The filters establish the analytical context, the KPI cards summarize the selected data, and the supporting visuals explain the factors behind those summary values.

Dashboard Component	Explanation
Date Filter	Allows the user to change the analysis period and examine the dashboard at annual or selected-period level.
Shipping Mode Filter	Allows the analysis to be restricted to Air, Rail, Road or Sea.
KPI Cards	Present the main numerical indicators such as total orders, total units, total order value, inventory turnover and days of inventory.
Category Analysis	Shows how order value is distributed across product categories.
Shipping Mode Analysis	Shows the contribution of different logistics modes to the overall order activity.
Audit Table	Provides detailed information for closer inspection of categories, organizations, buyers, modes and units.
Category Turnover Analysis	Compares category-level turnover using the documented analytical approach.
Units and Delay Timeline	Shows how ordered units and average delay vary over time.
Mode / Category Delay Analysis	Allows average delay to be compared across logistics modes and product categories.

These visuals are designed to work together. The KPI cards provide the high-level summary, while the charts and tables allow the user to investigate the composition behind those summary values. The interactive filters make it possible to move from an overall view to a selected month, shipping mode or category without creating separate reports.

4.3 Inventory and Performance Interpretation

In the documented full-year 2023 view, the dashboard reports approximately 1K total orders, 504K total units and $25M total order value. It also reports 4.00x inventory turnover and 91.25 days of inventory. The documentation specifies that the turnover and inventory-days figures are proxy-based indicators under the stated inventory assumption, so these values should be interpreted as analytical estimates rather than direct physical-stock measurements.

The annual view also presents a relatively distributed shipping-mode mix. The documented figures show Air at 26.3%, Road at 25.6%, Rail at 25.5% and Sea at 22.6%. This allows the user to understand how the overall order activity is distributed among the available transportation modes.

4.4 Time and Mode Analysis

The dashboard can be filtered to individual months to reveal changes that may not be visible in the annual summary. For example, the documented December 2023 view contains 82 orders, approximately 40K units and around $2M order value, whereas the October 2023 view contains 96 orders, approximately 48K units and around $3M order value. These views demonstrate how the same dashboard can be reused to examine different periods.

Separate shipping-mode views provide another level of analysis. The supplied documentation records Air with 263 orders and 137K units, Rail with 255 orders and 128K units, Road with 256 orders and 130K units, and Sea with 226 orders and 109K units. These values help illustrate the relative activity handled by each mode in the supplied data.

4.5 Milestone 1 Contribution

The main contribution of Milestone 1 is the establishment of a reliable analytical baseline. It introduces the dashboard structure, core dimensions, KPI approach, filtering mechanism and initial inventory and logistics indicators. The following milestones build on this foundation rather than replacing it.

Milestone 1 – Output Screenshots

The following screenshots document the main output views produced during Milestone 1.

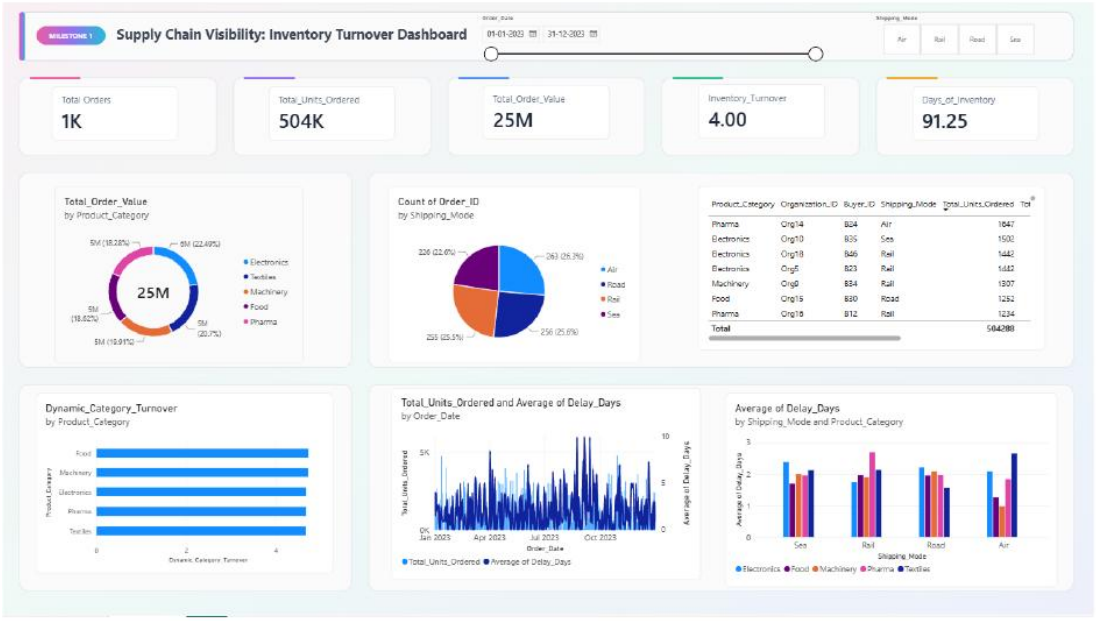


Figure 1. Milestone 1 – Overall 2023 dashboard

6. Milestone 2 – Dashboard Explanation

Inventory & Delivery Performance Analytics

Milestone 2 builds directly on the visibility created in Milestone 1 but changes the analytical emphasis toward delivery execution. The dashboard examines whether orders are being delivered on time, how severe delays are, how performance changes across periods and which suppliers or operational nodes are associated with different levels of delivery performance.

The dashboard is designed as a filter-responsive decision-support system. The same underlying model is used for the annual baseline, recent-period comparisons and shipping-mode analysis. As the user changes the date or logistics filters, the KPI cards and diagnostic visuals update within the same filter context.

6.1 Data Transformation and Derived Fields

Power Query is used as the ETL layer, while calculated fields provide additional analytical categories. Delivery_Status classifies records into On-Time and Delayed groups based on Delay_Days. Stock_Velocity_Segment separates Fast-Moving and Slow-Moving activity using the documented quantity threshold. A calendar dimension is also generated from the available order dates so that monthly trends can be displayed chronologically. filecite turn0file0 L66-L86

6.2 DAX Measures

The main delivery measures are created after the data model so that they respond to the current report filters. The measures include total deliveries, on-time deliveries, on-time delivery rate and average delay days. The supplied Milestone 2 documentation provides the following analytical logic. filecite turn0file0 L123-L140

```
Total_Deliveries = COUNTROWS('supply_chain_resilience_dataset csv')


On_Time_Deliveries =
CALCULATE(
    COUNTROWS('supply_chain_resilience_dataset csv'),
    'supply_chain_resilience_dataset csv'[Delivery_Status] = "On-Time"
)
```

```
On_Time_Delivery_Rate =
DIVIDE([On_Time_Deliveries], [Total_Deliveries], 0)

Avg_Delay_Days =
AVERAGE('supply_chain_resilience_dataset csv'[Delay_Days])
```

6.3 Dashboard Components

Component	Purpose
Date Filter	Controls the period used for delivery-performance analysis.
Logistics Mode Slicer	Allows Air, Rail, Road and Sea to be examined separately.
Total Deliveries Card	Shows the delivery workload for the selected context.
On-Time Delivery Rate Card	Shows the proportion of deliveries classified as On-Time.
Average Delay Card	Shows the average delay in days.
Total Units Fulfilled Card	Shows the volume of units associated with the selected period.
SLA Donut	Shows the distribution between On-Time and Delayed deliveries.
Monthly Trend	Shows how average delay changes over time.
Supplier Performance Chart	Compares on-time delivery performance between supplier or organization nodes.
Holding Risk and Demand Matrix	Connects demand velocity, volume and delay-related exposure.
Category-Mode Delay Treemap	Shows where delay is concentrated across product categories and shipping modes.

The visual architecture is documented as a coordinated set of components rather than isolated charts. The purpose is to allow the user to begin with a service-level KPI and then trace the result through time, supplier, category and transportation dimensions. filecite turn0file0 L143-L162

6.4 Performance Comparison

The full-year view reports approximately 1K deliveries, a displayed on-time rate of 0.49, an average delay of 1.94 days and 504K units. The recent-period views provide a more focused comparison: the latest two-month view contains 184 deliveries, a 0.49 displayed on-time rate, 2.20 average delay days and 95K units, while the previous two-month view contains 188 deliveries, a 0.41 displayed on-time rate, 2.26 average delay days and 93K units. filecite turn0file0 L189-L199 filecite turn0file0 L206-L234

This comparison demonstrates why the dashboard includes both a baseline view and recent-period filtering. A user can first understand the overall delivery pattern and then examine whether the most recent period differs in workload, on-time performance or delay severity.

6.5 Milestone 2 Contribution

Milestone 2 adds the service-performance layer to the project. It changes the main analytical question from inventory and demand visibility to delivery execution. This makes the overall system capable of examining both what volume is moving through the supply chain and how successfully that volume is being delivered.

Milestone 2 – Output Screenshots

The following screenshots show the major output states documented for Milestone 2.

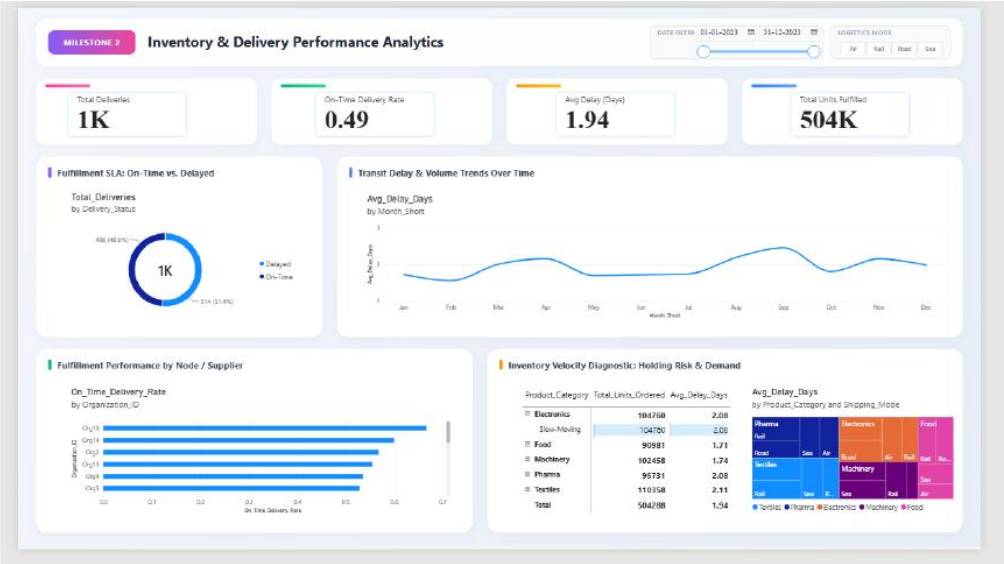


Figure 2. Milestone 2 – Overall full-year dashboard

7. Milestone 3 – Dashboard Explanation

Transportation & Supplier Performance Analytics

Milestone 3 extends the project into transportation expenditure, supplier performance and disruption-risk analysis. The dashboard is designed to answer a different set of operational questions: how much estimated transportation cost is being incurred, what the average shipment cost is, how suppliers compare in volume and reliability, and which disruption categories are associated with delay and historical disruption activity.

7.1 DAX Measures

Since the dataset does not provide a dedicated freight-expense column, the dashboard estimates transportation cost from Quantity_Ordered and a shipping-mode-specific freight rate. The rates supplied for this milestone are Air 18.5, Sea 4.2, Rail 6.8 and Road 9.5, with 8.0 used as the default rate.

```
Total_Transportation_Cost =
SUMX(
    'supply_chain_resilience_dataset csv',
    'supply_chain_resilience_dataset csv'[Quantity_Ordered] *
    SWITCH(
        'supply_chain_resilience_dataset csv'[Shipping_Mode],
        "Air", 18.5,
        "Sea", 4.2,
        "Rail", 6.8,
        "Road", 9.5,
        8.0
    )
)
Avg_Shipment_Cost =
DIVIDE(
    [Total_Transportation_Cost],
    COUNTROWS('supply_chain_resilience_dataset csv'),
    0
)
Order_Fulfillment_Rate =
DIVIDE(
    CALCULATE(
        COUNTROWS('supply_chain_resilience_dataset csv'),
        'supply_chain_resilience_dataset csv'[Delay_Days] = 0,
        'supply_chain_resilience_dataset csv'[Supply_Risk_Flag] = 0
    ),
    COUNTROWS('supply_chain_resilience_dataset csv'),
```

```
    0
)
On_Time_Delivery_Rate =
DIVIDE(
    CALCULATE(
        COUNTROWS('supply_chain_resilience_dataset csv'),
        'supply_chain_resilience_dataset csv'[Delay_Days] <= 0
    ),
    COUNTROWS('supply_chain_resilience_dataset csv'),
    0
)
Supplier_Quality_Score =
AVERAGE(
    'supply_chain_resilience_dataset csv'[Supplier_Reliability_Score]
)


Total_Shipments =
COUNTROWS('supply_chain_resilience_dataset csv')
```

7.2 Interactive Slicers

Four slicers are placed in the dashboard header so that the complete report can be explored through different operational dimensions. Supplier_ID filters supplier-specific activity, Shipping_Mode isolates a transportation mode, Product_Category focuses the analysis on a product group, and Disruption_Type isolates a specific disruption class.

Slicer	Field	Style
Supplier	Supplier_ID	Dropdown
Carrier / Mode	Shipping_Mode	Dropdown
Category / Risk	Product_Category	Dropdown
Disruption Type	Disruption_Type	Dropdown

7.3 KPI Cards

The four KPI cards provide an immediate summary of the selected dashboard context. Total Transportation Cost represents the estimated freight expenditure under the specified rate assumptions. Average Shipment Cost represents the estimated transportation cost divided by the number of transactions. Order Fulfillment Rate uses the defined zero-delay and zero-risk condition, while On-Time Delivery Rate uses the Delay_Days ≤ 0 condition.

KPI	Measure	Formatting
Total Transportation Cost	[Total_Transportation_Cost]	Currency, 0 decimals
Average Shipment Cost	[Avg_Shipment_Cost]	Currency, 2 decimals
Order Fulfillment Rate	[Order_Fulfillment_Rate]	Percentage, 1 decimal
On-Time Delivery Rate	[On_Time_Delivery_Rate]	Percentage, 1 decimal

7.4 Middle Visuals

The Transportation Cost Hierarchy is implemented as a clustered column chart with Product_Category, Shipping_Mode and Supplier_ID placed in sequence on the axis. This creates an interactive drill-down path from a broad category view to individual supplier-level transportation cost.

The Carrier-wise Transportation Cost and Mode Efficiency visual uses Shipping_Mode as the legend and Total_Transportation_Cost as the value. The On-Time Delivery Rate is provided as supporting tooltip information, allowing transportation expenditure to be viewed alongside delivery performance.

7.5 Bottom Visuals

The Supplier-wise Shipment Volume chart ranks suppliers by shipment activity and uses a Top 10 filter to keep the comparison focused. The Supplier Quality and Reliability Score chart provides a complementary view by comparing Supplier_Reliability_Score across suppliers. Together, these two visuals distinguish shipment scale from supplier quality.

The Disruption and Delay Risk Matrix combines Disruption_Type and Shipping_Mode as row levels. It includes total shipments, average delay, historical disruption count and supplier quality. This makes it possible to examine the relationship between disruption exposure and transportation performance within one visual.

7.6 Dashboard Results

In the displayed dashboard state, the headline values are approximately 2.53M total transportation cost, 9.64K average shipment cost, 0.52 order fulfillment rate and 1.76 for the displayed delay/transit indicator. These values represent the current filter context and will change when a supplier, mode, category or disruption type is selected.

The dashboard therefore moves beyond simple cost reporting. The user can begin with the total transportation cost, identify the shipping mode contributing to that cost, drill down to suppliers, compare supplier reliability and then use the risk matrix to investigate disruption and delay patterns.

7.7 Milestone 3 Contribution

Milestone 3 completes the three-stage progression by adding transportation economics and supplier-risk analysis to the visibility and delivery layers developed earlier. It provides a broader operational view in which cost, volume, service quality and disruption exposure can be examined together.

Milestone 3 – Output Screenshots

The following screenshot represents the final Milestone 3 dashboard output.

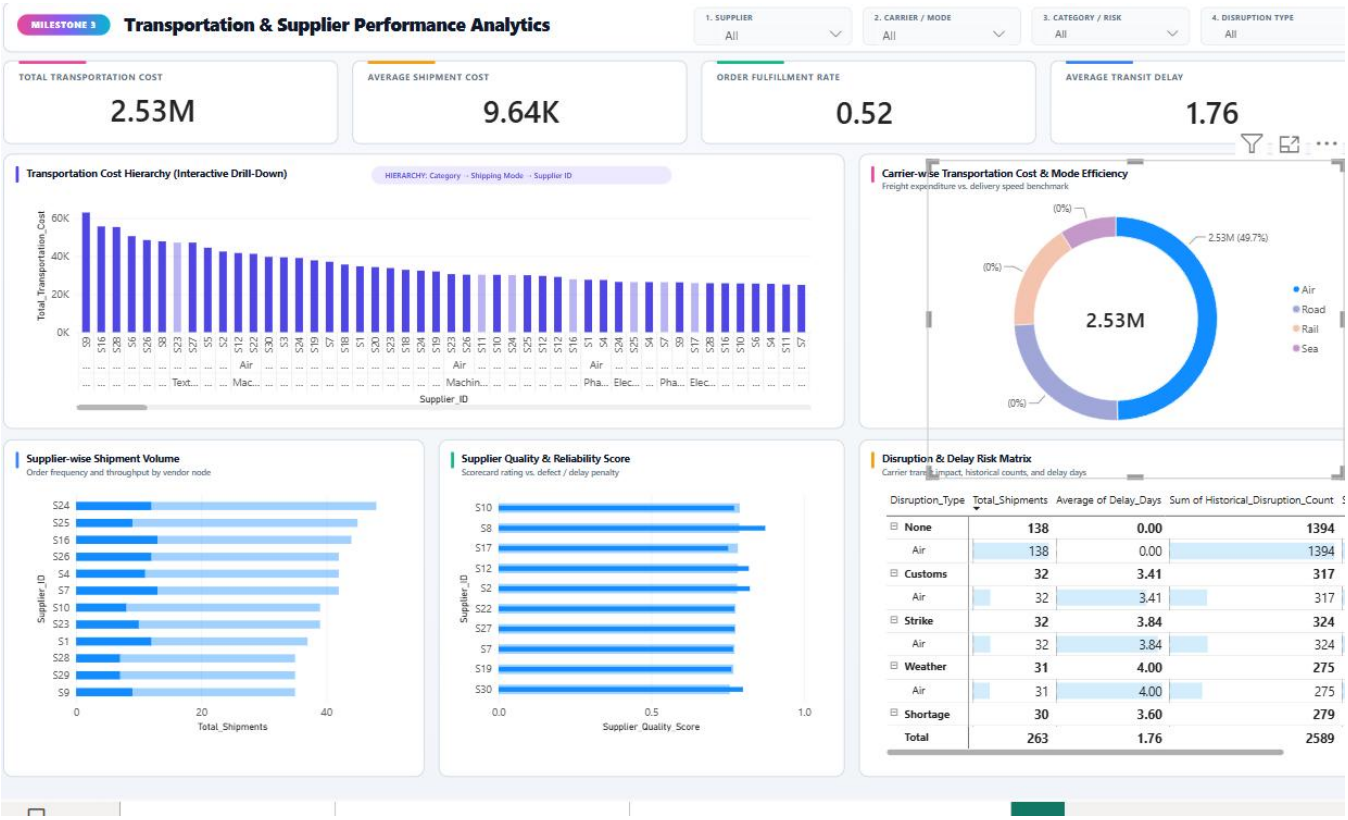


Figure 3. Milestone 3 – Transportation & Supplier Performance Analytics dashboard

8. Milestone 4 – Warehouse Analytics & Performance Optimization Dashboard

8.1 Milestone 4 Introduction

Milestone 4 extends the Supply Chain Visibility System from order, delivery, transportation, and supplier-level analysis into facility-level warehouse analytics. The milestone introduces a physical fulfillment layer that connects shipping activity with warehouse identity, storage capacity, operational expenditure, picking accuracy, and delivery service performance. The resulting reporting suite is designed to help users examine how individual fulfillment facilities handle order volume and inventory throughput while also identifying operational bottlenecks and service-level gaps.

The milestone is implemented as a three-page Power BI reporting suite consisting of the Executive Dashboard, Warehouse Efficiency Report, and Performance Optimization Dashboard. The pages are connected through the report navigation controls shown at the bottom of the dashboards, allowing the user to move between network-level performance, facility efficiency, and detailed optimization analysis. Together, the three pages provide both a management summary and a high-density diagnostic view of warehouse operations.

8.2 Data Cleaning & Transformation

Power Query transformations were applied to prepare the existing supply-chain transaction data for warehouse-level analysis while preserving the underlying transaction records. A custom route-based warehouse key was introduced so that each transaction could be associated with a fulfillment facility. The Warehouse_ID field was assigned according to the shipping mode using the specified mapping: Air to WH-1, Road to WH-2, Rail to WH-3, and Sea to WH-4. This transformation provides a consistent facility identifier that can be used to establish the warehouse relationship in the analytical model.

Header sanitization was also performed to remove whitespace artifacts and tab delimiters such as #(tab) from imported field definitions, including Warehouse_ID and Region. Explicit data types were then enforced across the warehouse attributes so that identifiers are treated as text, capacity and count fields use integer types, and performance rates are represented as decimal values. These transformations provide a consistent foundation for the warehouse dimension and the measures used throughout the three dashboards.

8.3 Data Modelling & Star Schema Extension

The existing star-schema model was extended through the creation of a dedicated Dim_Warehouse dimension table. The dimension contains Warehouse_ID, Warehouse_Name, Region, Storage_Capacity, Base_Op_Cost, and Target_Picking_Acc. These attributes provide the descriptive and operational context required to evaluate warehouse utilization, cost, picking quality, and delivery performance.

An active one-to-many (1:*) relationship was configured between Dim_Warehouse[Warehouse_ID] and Fact_SupplyChain_Orders[Warehouse_ID], with single-direction cross-filtering. This keeps the warehouse dimension integrated with the fact table without introducing circular dependencies. As a result, warehouse-level selections can filter the transaction data consistently across the report pages while retaining the structure of the original analytical model.

8.4 KPI Creation (DAX Measures)

A set of DAX measures was created to quantify warehouse workload, capacity utilization, service performance, accuracy, delay, and operating expenditure. The measures are reused across the report pages so that the same definitions are applied when users move from the executive summary to facility diagnostics.

Measure	Calculation / Definition	Purpose
Total_Orders	DISTINCTCOUNT(Order_ID)	Counts distinct supply-chain orders.
Sum_Total_Units_Ordered	SUM(Quantity_Ordered)	Calculates the total quantity ordered.
Warehouse_Utilization_Rate	DIVIDE([Sum_Total_Units_Ordered], SUM(Dim_Warehouse[Storage_Capacity]), 0)	Compares ordered-unit volume with warehouse storage capacity.
On_Time_Delivery_Rate	DIVIDE(CALCULATE(COUNTROWS(...), Delay_Days <= 0), COUNTROWS(...), 0)	Measures the share of transactions delivered on time or early.
Picking_Accuracy_Rate	Performance measure based on shortages	Evaluates picking accuracy after

	and customs-delay disruptions	accounting for the specified disruption conditions.
Order_Accuracy_Rate	Non-risk orders / total shipments	Measures the proportion of shipments not classified as risk orders.
Average_Delay_Days	AVERAGE(Delay_Days)	Calculates the average delivery delay.
Total_Operations_Cost	Facility baseline + per-unit handling + machine energy expenditure	Aggregates the specified warehouse operating-cost components.

8.5 Milestone 4 Dashboard Explanation

8.5.1 Executive Dashboard

The Executive Dashboard provides the management-level view of warehouse and network performance. Its KPI ribbon summarizes Total Orders, Sum Total Units Ordered, Total Operations Cost, On-Time Delivery Rate, and Order Accuracy Rate. These indicators give the user a compact view of workload, operating expenditure, and service quality before moving into more detailed visuals.

The central Warehouse Volume Handled vs. Operations Cost visual compares the quantity processed by each facility with its associated operational cost, making it possible to inspect the relationship between workload and expenditure. A regional donut chart presents the distribution of total orders across warehouse regions. The Executive Warehouse Summary & SLA Compliance Matrix then combines warehouse-level order volume, units ordered, operations cost, and on-time delivery rate for direct facility comparison.

8.5.2 Warehouse Efficiency Report

The Warehouse Efficiency Report focuses on how effectively individual facilities use their available storage and process incoming workload. The dashboard includes KPI indicators for warehouse storage utilization, picking accuracy, total operations cost, and average delay. These indicators are complemented by a monthly warehouse inventory throughput and flow-velocity trend, which shows how unit volume changes across the reporting period.

The Storage Space Allocation by Facility Hub donut chart summarizes the distribution of the reported storage capacity across the warehouse network. The Stock On-Hand vs. Capacity Limit visual provides a facility-level comparison between stock volume and capacity limits, while the Picking & Fulfillment Accuracy by Category visual examines category-specific accuracy. The Warehouse Operational Bottleneck Matrix brings these indicators together with total orders, average delay, and total operational cost for facility-level diagnosis.

8.5.3 Performance Optimization Dashboard

The Performance Optimization Dashboard is designed as a detailed diagnostic view rather than a high-level summary. Its main component is a warehouse performance matrix that consolidates facility workload and operational indicators in one place. The matrix includes Warehouse Name, Total Orders Handled, Sum Total Units Ordered, Warehouse Utilization Rate, On-Time Delivery Rate, Picking Accuracy Rate, and additional operational attributes available in the model.

The matrix supports multi-attribute performance analysis by allowing users to compare facilities across volume, capacity usage, delivery service, and picking quality. The displayed results show the four modeled facilities—Central Air Hub, Metro Road Depot, Rail Terminal East, and Maritime Port West—along with a total row for the network. This layout makes the page suitable for identifying facilities that require closer investigation based on their combination of workload, utilization, service performance, and accuracy measures.

Milestone 4 Screenshots

The following screenshots document the three Power BI pages developed for Milestone 4. They are presented in the report in the same functional order as the reporting suite: executive overview, warehouse efficiency analysis, and detailed performance optimization.

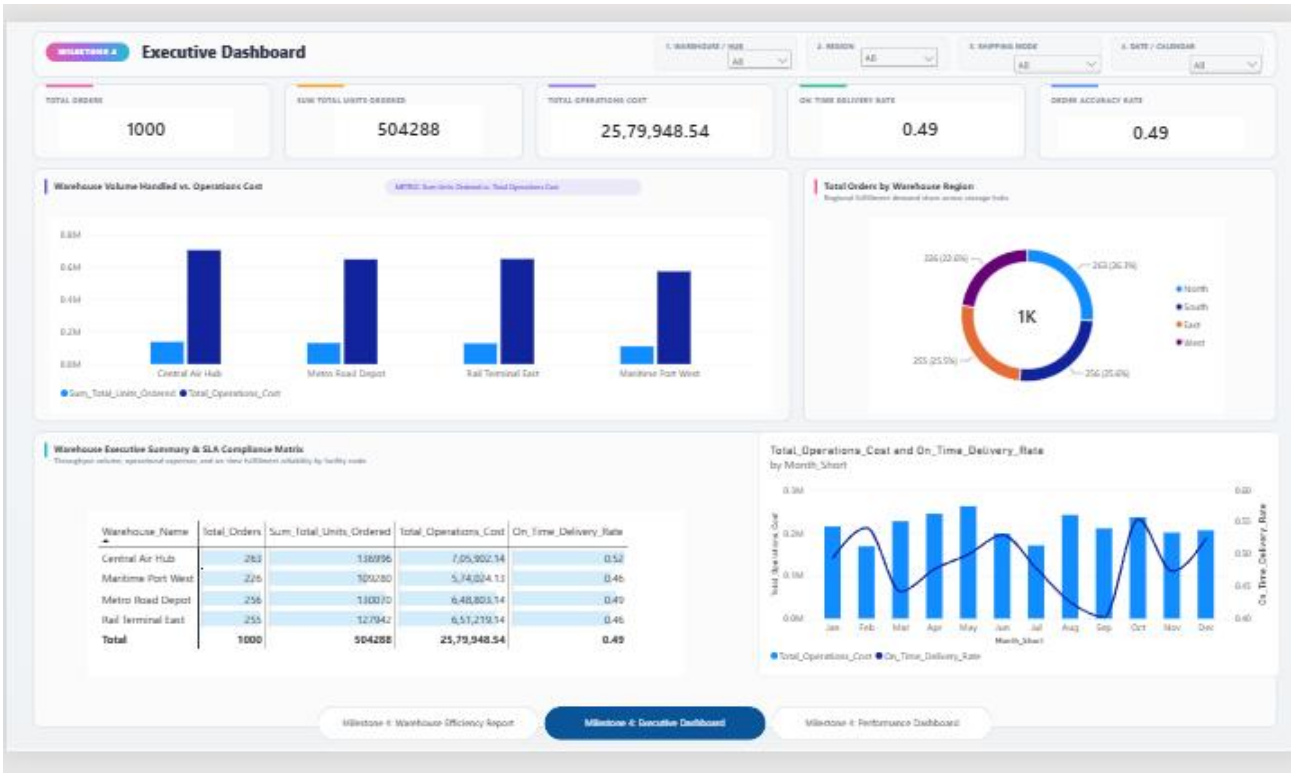


Figure 12.1. Milestone 4 – Executive Dashboard

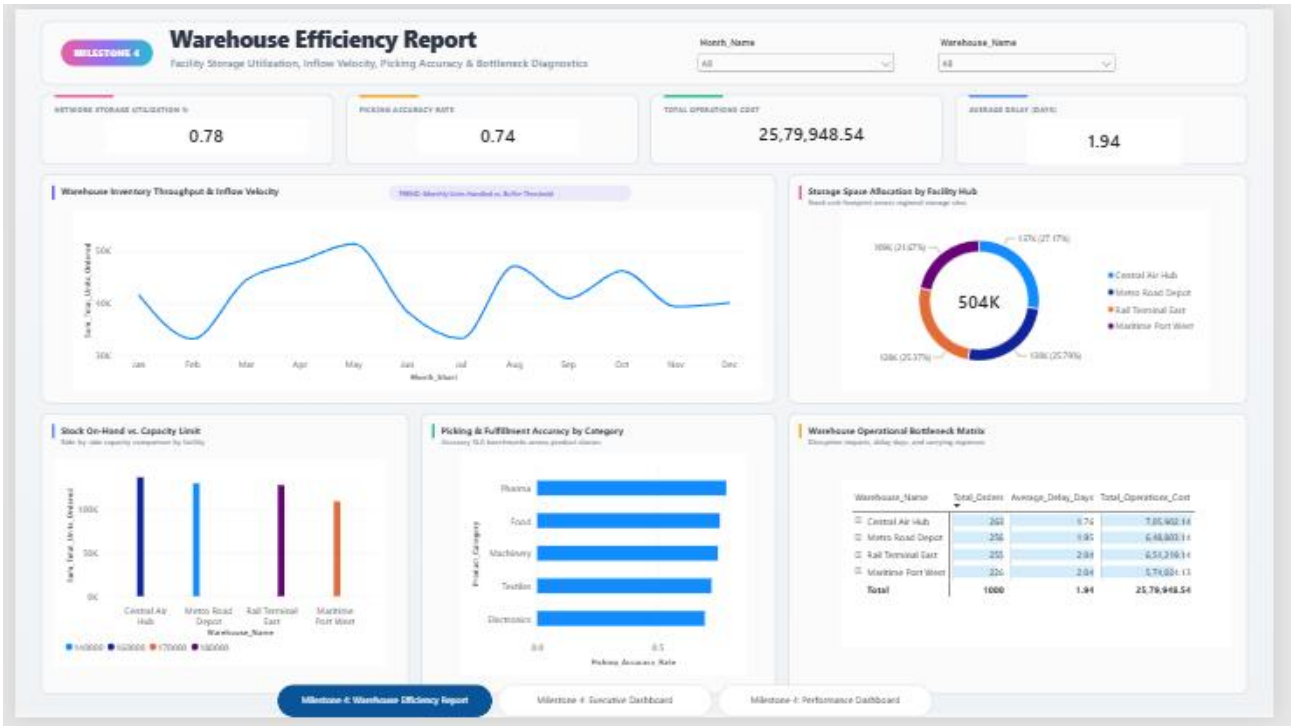


Figure 12.2. Milestone 4 – Warehouse Efficiency Report

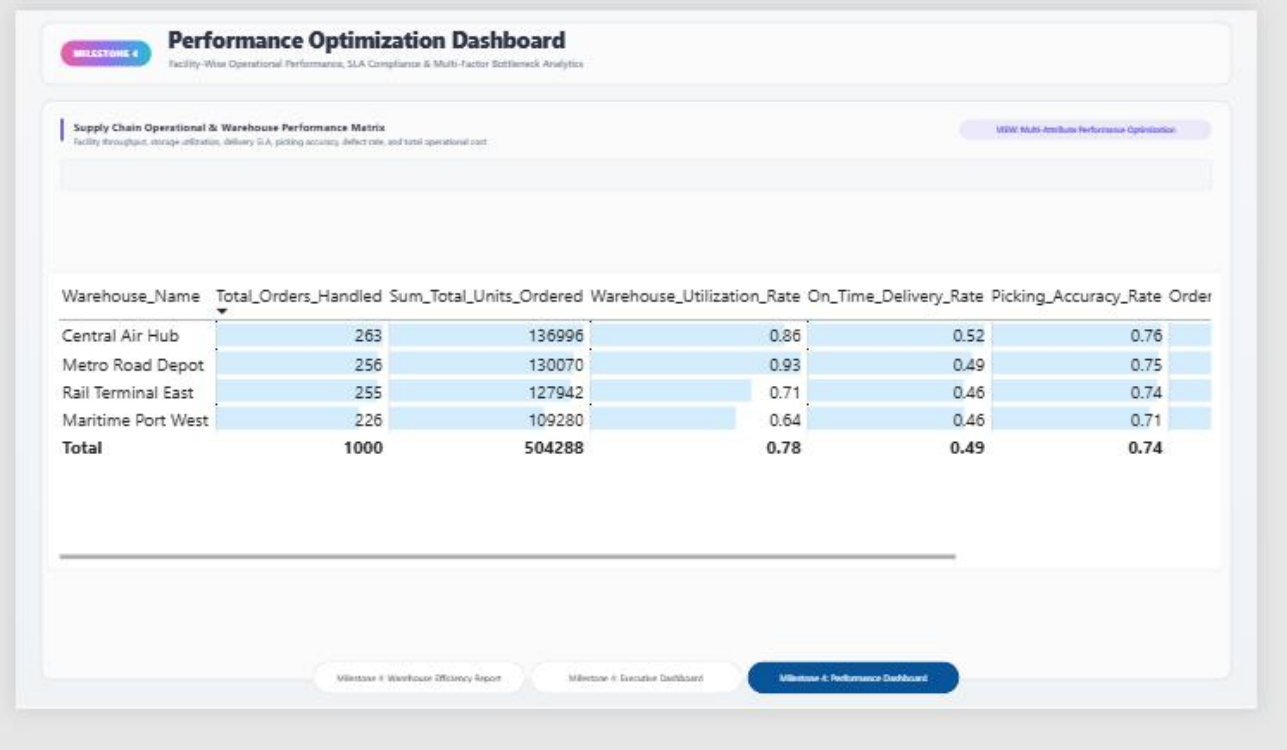


Figure 12.3. Milestone 4 – Performance Optimization Dashboard

8.7 Milestone 4 Outcome

Milestone 4 completes the physical-facility layer of the Supply Chain Visibility System by linking transaction activity to warehouse identity, capacity, operational cost, service performance, and accuracy. The three dashboards provide progressive levels of analysis: the Executive Dashboard summarizes the network, the Warehouse Efficiency Report examines facility utilization and throughput, and the Performance Optimization Dashboard consolidates detailed warehouse indicators for comparison and diagnosis. This structure extends the earlier milestones into a facility-oriented optimization view without changing the underlying transaction records.

The page-navigation controls visible at the bottom of the dashboards provide a consistent transition between the three Milestone 4 views. This supports a workflow in which users can begin with executive indicators, investigate warehouse efficiency, and then move to the detailed performance matrix for facility-level analysis.

9. Conclusion

The Supply Chain Visibility System with Optimization Analytics provides a structured Power BI environment for examining supply-chain operations from multiple perspectives. The project begins with inventory and demand visibility, extends into delivery and service-level performance, and finally incorporates transportation expenditure, supplier quality and disruption risk.

The use of interactive slicers, drill-down hierarchies, KPI cards, comparative charts and risk matrices allows the dashboard to support both high-level monitoring and detailed investigation. Instead of relying on a single summary number, users can move through the different analytical layers to understand the operational context behind the result.

Overall, the three milestones form a connected analytical system in which each stage adds a distinct purpose while reusing the underlying data and dashboard structure. This creates a more complete view of supply-chain performance covering inventory, delivery, transportation, supplier reliability and disruption-related risk.

10. References

1. **Infosys Springboard – Learning Microsoft Power BI.**
   Infosys Springboard, Infosys Limited. Used as a primary learning reference for Microsoft Power BI, including Power BI concepts, report development and analytical workflows.
   Learning Microsoft Power BI – Infosys Springboard
2. **Infosys Springboard – Hands-On Data Visualization with Microsoft Power BI.**
   Infosys Springboard, Infosys Limited. Used as a practical reference for data visualization, interactive reports, dashboards and visual analytics in Power BI.
   Hands-On Data Visualization with Microsoft Power BI – Infosys Springboard
3. **Infosys Springboard – Power BI for Business Professionals.**
   Infosys Springboard, Infosys Limited. Used as a supporting reference for applying Power BI to business-oriented analytics, reporting and decision-making.
   Power BI for Business Professionals – Infosys Springboard
4. **Microsoft Learn – Power BI Documentation.** *Power BI documentation: Data modelling, reports, dashboards and analytics.*
   Microsoft Power BI Documentation
5. **Microsoft Learn – Star Schema and Power BI.** *Understand star schema and its importance for Power BI.*
   Microsoft Learn – Star Schema Guidance
6. **Microsoft Learn – Model Relationships in Power BI Desktop.** *Model relationships in Power BI Desktop.*
   Microsoft Learn – Model Relationships
7. **Microsoft Learn – Create and Manage Relationships in Power BI Desktop.** Supports relationship configuration, cardinality and cross-filter direction in Power BI models.
   Microsoft Learn – Create and Manage Relationships
8. **Microsoft Learn – DAX Reference.** *Data Analysis Expressions (DAX) Reference.* Supports the DAX measures and functions used in the project.
   Microsoft DAX Reference
9. **Microsoft Learn – Semantic Model Designer Documentation.** Supports data transformation, relationships, measures, calculated columns and semantic-model development.
   Microsoft Learn – Semantic Model Designer
10. **Microsoft Learn – Dimensional Model to Power BI Report.** Supports dimensional modelling and development of analytical reports from a structured data model.
    Microsoft Learn – Dimensional Model to Power BI Report
11. **Microsoft Learn – Active vs. Inactive Relationship Guidance.** Supports understanding active relationships and filter propagation within Power BI models.
    Microsoft Learn – Active vs. Inactive Relationships
12. **Kaggle – Accenture Supply Chain Resilience Challenge Dataset.** Used as the reference for the transactional supply-chain dataset and its operational fields used for analysis.
    Kaggle – Accenture Supply Chain Resilience Challenge